



In Search of the Computational Primitives of Language

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Computation and Theory Building

*[...] this is a confusion of two quite separate issues, **simulation and explanation**. [...] What we are **really** interested in [...] is explanation — in developing models that help us **understand how it is that people behave** that way, not merely demonstrating that we can build an artifact that behaves similarly.*

(Kaplan, 1995)

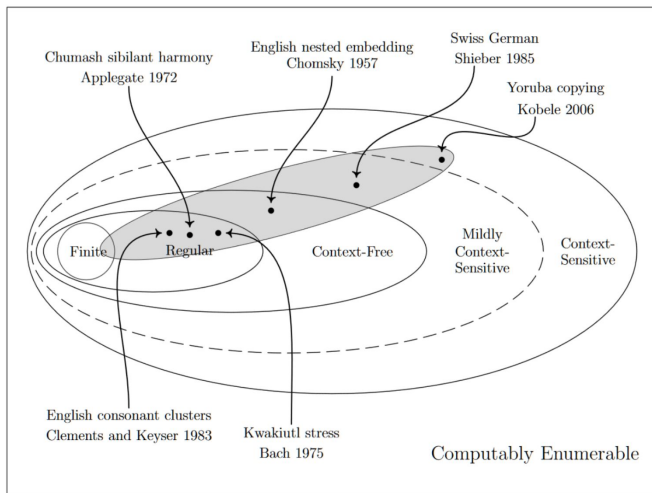
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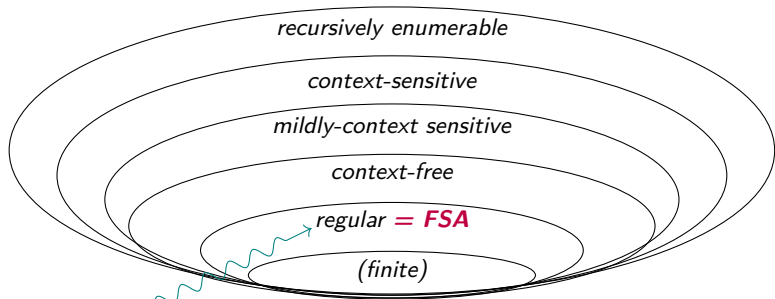
(Kaplan, 1995)

Interpretability for the win!

A Lens: Formal Language Theory



Spoken Languages' Phonology as a Regular System



Phonology

Kaplan and Kay (1994)

Local Phonotactic Dependencies

1 Intervocalic voicing in Italian

Forbid voiceless segments in between two vowels

- (1) a. * /kasa/
b. /kaza/
→ cf. orthography: “*casa*”

Intervocalic voicing is Strictly Local (SL)

- ▶ Forbid voiceless segments in-between two vowels: *V[-voice]V
- ▶ **Italian:** *ase, *ise, *ese, *isi, ...

\$ k a s a \$

\$ k a z a \$

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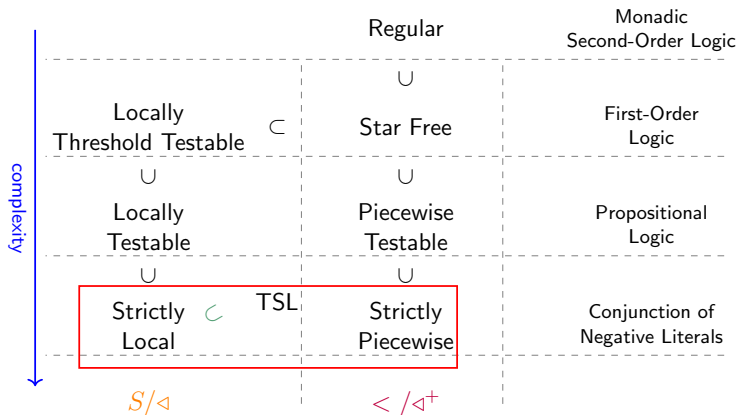
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Beyond Automata: Subregular Languages¹



¹McNaughton & Papert (1976), Heinz (2011), Chandlee & Heinz (2014), De Santo & Graf (2019), De Santo & Rawski (2022), a.o.

So What?

- ▶ Capture minimal information requirements
(McMullin 2016, De Santo & Rawski 2022)
- ▶ Provably correct learning algorithms
(De Santo & Aksenova 2021, Johnson & De Santo 2023,2024)
- ▶ Predictions for human learning
(Lai 2015, Avcu & Hestevic 2021, De Santo & Rawski 2022)

So What?

- ▶ Capture minimal information requirements
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⇒ Architectural Constraints?
- ▶ Provably correct learning algorithms
(De Santo & Aksenova 2021, Johnson & De Santo 2023,2024)
⇒ Interpretability of datasets
- ▶ Predictions for human learning
(Lai 2015, Avcu & Hestevic 2021, De Santo & Rawski 2022)
⇒ Behavioral benchmarking

FLT, Linguistics, and LLM Expressivity

Saturated Transformers are Constant-Depth Threshold Circuits

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Between Circuits and Chomsky: Pre-pretraining on Formal Languages Imparts Linguistic Biases

Michael Y. Hu¹ Jackson Petty² Chuan Shi¹ William Merrill¹ Tal Linzen^{1,2}

Neuro-Symbolic Language Modeling with Automaton-augmented Retrieval

Uri Alon¹ Frank F. Xu¹ Junxian He¹ Sudipta Sengupta² Dan Roth³ Graham Neubig¹
¹Language Technologies Institute, Carnegie Mellon University ²Amazon AWS ³AWS AI Labs
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What Makes Instruction Learning Hard? An Investigation and a New Challenge in a Synthetic Environment

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Switching Gears: Benchmarking with Sentence Processing

Not All Structures Are Processed Equally

- Subject VS object relative clause

SRC **The horse** [_{RC} that kicked the wolf] went home.

ORC **The horse** [_{RC} that the wolf kicked] went home.

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Assessing the Ability of LSTMs to Learn Syntax-Sensitive Dependencies

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SRC **The horse** [_{RC} that **t** kicked the wolf] went home.

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- ▶ Attachment preferences

1.a I shot an elephant in my pajamas

1.b I [shot an elephant] [in my pajamas]

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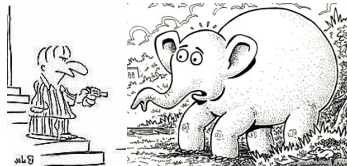
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Ambiguity All the Way Down



So What?

Ambiguity is ubiquitous in natural language!

For Cognitive Science

- ▶ How do humans handle multiple structural representations?
- ▶ What principles guide ambiguity resolution cross-linguistically?
- ▶ Language specific properties vs. general biases/mechanisms?

For NLP

- ▶ How do LLMs handle multiple structural representations?
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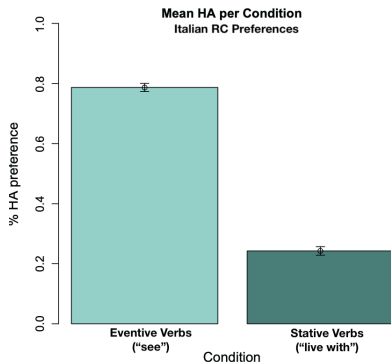
Ambiguity and Relative Clauses Cross-linguistically ²

- ▶ They saw the daughter of the actress that was on the balcony
- | | | | |
|-----------------|---------------------|--------------------|----|
| NP ₁ | The daughter | was on the balcony | HA |
| NP ₂ | The actress | was on the balcony | LA |

²Grillo & Costa (2015,) De Santo & Lee (2023), Lee & De Santo (2024)

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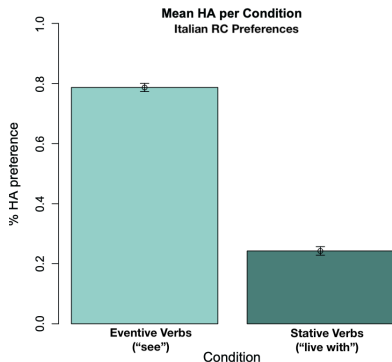
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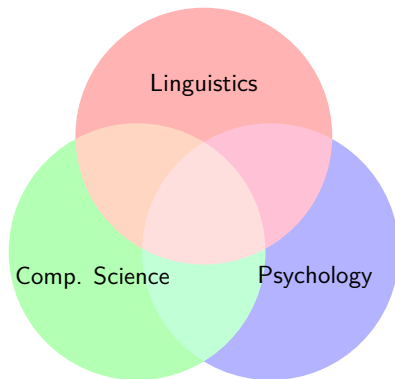


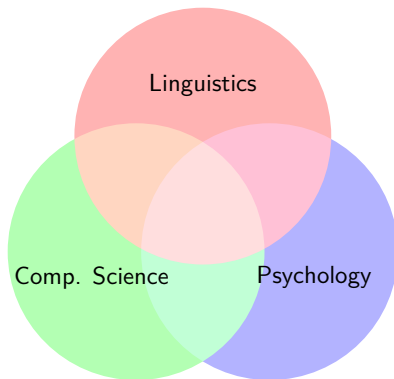
LLMs and Italian RCs?



Check out the poster!

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Let's chat!

Unbounded Dependencies Are Not SL

► **Samala Sibilant Harmony**

Sibilants must not disagree in anteriority.

(?)

- (2) a. * ha **s** x i n t i l a w a **ʃ**
- b. * ha **ʃ** x i n t i l a w a **s**
- c. ha **ʃ** x i n t i l a w a **ʃ**

Example: Samala

* \$ h a **s** x i n t i l a w a **ʃ** \$

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- (2) a. * ha^sxintilawa^f
b. * ha^fxintilawa^s
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* \$ h a ^s x i n t i l a w a ^f \$

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Example: Samala

*\$ ha^sxintilawa^ʃ\$
\$ ha^ʃxintilawa^ʃ\$

► **But:** Sibilants can be arbitrarily far away from each other!

*\$ ^st a j a n o w o n w a ^ʃ\$

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* \$ ^s t a j a n o w o n w a ^f \$

Unbounded Dependencies are TSL

- ▶ Let's revisit Samala Sibilant Harmony

- (3) a. * ha^sxintilawa^ʃ
b. * ha^ʃxintilawa^s
c. ha^ʃxintilawa^ʃ

- ▶ What do we need to project? [+strident]
- ▶ What do we need to ban? *[+ant][−ant], *[−ant][+ant]

I.E. *^sʃ, *^sʒ, *^ʒʃ, *^ʒʒ, *^ʃs, *^ʒs, *^ʃz, *^ʒz

Example: TSL Samala

^s ʃ

.....

* \$ha^sxintilaw^ʃ\$

ʃ ʃ

.....

^{ok} \$ha^ʃxintilaw^ʃ\$

Unbounded Dependencies are TSL

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Example: TSL Samala


* \$ha^sxintilawa^ʃ\$


ok \$ha^ʃxintilawa^ʃ\$